

RMSNorm CUDA Kernel Benchmark Report

Configuration

Parameter	Value
Kernel	rmsnorm_fwd_kernel
Precision	BF16(__nv_bfloat16)
GPU Architecture	NVIDIA Ada Lovelace
Compute Capability	sm_89
Input Shape	[1024 x 2048]
Hidden Dimension	2048
Rows	1024
Threads per Block	256
Blocks	1024
Compiler	nvcc -arch=sm_89 -lineinfo -O3
Reduction	Warp shuffle + shared-memory block reduction
Vectorization	uint4 + __nv_bfloat162
Profiling Tool	NVIDIA Nsight Compute

Performance Benchmark

Metric	Value
Average Kernel Execution Time	0.1011 ms
Nsight Compute Measured Duration	22.46 µs
Number of Rows Processed	1024
Hidden Size	2048
Total Elements Processed	2,097,152 BF16 values

Metric	Value
Input Size	~4 MB
Output Size	~4 MB
Weight Size	~4 KB
Maximum Absolute Error	0.00385129
Numerical Validation	Passed

Nsight Compute Profiling Results

Category	Metric	Value
Kernel Launch	Grid	(1024, 1, 1)
Kernel Launch	Block	(256, 1, 1)
Kernel Launch	Registers / Block Limit	8 blocks
GPU	Compute Capability	8.9
GPU Frequency	DRAM Frequency	6.99 GHz
GPU Frequency	SM Frequency	1.89 GHz
Execution	Elapsed Cycles	42,400 cycles
Execution	Kernel Duration	22.46 μs
Throughput	Memory Throughput	187.12 GB/s
Throughput	DRAM Throughput	83.70 %
Throughput	Compute Throughput	25.13 %
Throughput	L1/TEX Throughput	30.50 %
Throughput	L2 Throughput	25.07 %
Memory	Maximum Bandwidth Utilization	83.70 %
Memory	Memory Busy	25.02 %
Memory	Memory Pipes Busy	22.37 %
Cache	L1 Hit Rate	41.47 %
Cache	L2 Hit Rate	56.42 %
Cache	L2 Persisting Size	6.29 MB

Category	Metric	Value
Cache	L2 Compression	Disabled
Memory Efficiency	Local Memory Spilling	0
Memory Efficiency	Shared Memory Spilling	0

Warp and Scheduler Analysis

Metric	Value
Active Warps per Scheduler	10.67
Eligible Warps per Scheduler	0.44
Issued Warp per Scheduler	0.27
No Eligible Scheduler Cycles	73.43 %
Warp Cycles per Issued Instruction	40.17 cycles
Warp Cycles per Executed Instruction	40.39 cycles
Average Active Threads per Warp	32
Average Non-Predicated Threads per Warp	28.91

Occupancy Analysis

Metric	Value
Theoretical Active Warps per SM	48
Theoretical Occupancy	100 %
Achieved Occupancy	89.84 %
Achieved Active Warps per SM	43.12
Block Limit (SM)	24
Block Limit (Registers)	8
Block Limit (Shared Memory)	14
Block Limit (Warps)	6

Optimization Analysis

Issue	Observation
Primary Bottleneck	Memory bandwidth
DRAM Utilization	83.70 %
Compute Utilization	25.13 %
Register Spilling	None detected
Shared Memory Spilling	None detected
Occupancy	High (~90%)
Main Stall Reason	L1/TEX dependency stalls
L1/TEX Stall Contribution	~72.8% of warp stall cycles
Estimated Scheduler Improvement	16.3%

Final Result Summary

Metric	Result
Correctness	 Passed
BF16 Accuracy	Error = 0.00385129
Kernel Runtime	0.1011 ms
Achieved Occupancy	89.84%
Memory Bandwidth Utilization	83.70%
Register Spills	None
Shared Memory Spills	None
Performance Limitation	Memory access latency / L1TEX dependency stalls